CRM Database Design and SQL Script

This document gives you a starter CRM database for the `agentic_db` database on your SQL Server (AYAN-RIYAN). It covers the table design, what each field means, a full script to run in SSMS, sample data, and how it connects to your MCP server.

Part 1: The big picture

A CRM tracks the journey from a stranger to a customer. Five core tables cover it:

- **Accounts**: companies you sell to.
- **Contacts**: people who work at those accounts.
- **Leads**: unqualified prospects, before they become accounts and contacts.
- **Opportunities**: potential deals, linked to an account.
- **Activities**: calls, emails, meetings, and tasks linked to any of the above.

The flow is: Lead is created, then qualified, then converted into an Account + Contact + Opportunity, then the Opportunity moves through stages until Closed Won or Closed Lost.

Part 2: Table design

Accounts

Field	Type	Meaning
AccountId	INT, identity, primary key	Unique id
AccountName	NVARCHAR(200), required	Company name
Industry	NVARCHAR(100)	e.g. Technology, Retail
Website	NVARCHAR(200)	Company website
Phone	NVARCHAR(30)	Main phone
City, Country	NVARCHAR(100)	Location
AnnualRevenue	DECIMAL(18,2)	Company revenue
CreatedAt, UpdatedAt	DATETIME2	Audit timestamps

Contacts

Field	Type	Meaning
ContactId	INT, identity, primary key	Unique id
AccountId	INT, foreign key to Accounts	Which company they work at
FirstName, LastName	NVARCHAR(100)	Name
Email	NVARCHAR(200), unique	Email
Phone	NVARCHAR(30)	Phone
JobTitle	NVARCHAR(100)	Role
CreatedAt, UpdatedAt	DATETIME2	Audit timestamps

Leads

Field	Type	Meaning
LeadId	INT, identity, primary key	Unique id
FirstName, LastName	NVARCHAR(100)	Name
Company	NVARCHAR(200)	Their company name (text, not yet an account)
Email	NVARCHAR(200)	Email
Phone	NVARCHAR(30)	Phone
LeadSource	NVARCHAR(50)	Web, Referral, Event, Cold Call
Status	NVARCHAR(30)	New, Contacted, Qualified, Unqualified, Converted
IsConverted	BIT	1 once converted
ConvertedAccountId, ConvertedContactId, ConvertedOpportunityId	INT, nullable	Filled on conversion
CreatedAt, UpdatedAt	DATETIME2	Audit timestamps

Opportunities

Field	Type	Meaning
OpportunityId	INT, identity, primary key	Unique id
AccountId	INT, foreign key to Accounts	Which company
ContactId	INT, nullable, foreign key to Contacts	Primary contact
OpportunityName	NVARCHAR(200)	Deal name

Field	Type	Meaning
Stage	NVARCHAR(30)	Prospecting, Qualification, Proposal, Negotiation, Closed Won, Closed Lost
Amount	DECIMAL(18,2)	Deal value
Probability	INT	Win chance, 0 to 100
CloseDate	DATE	Expected close date
CreatedAt, UpdatedAt	DATETIME2	Audit timestamps

Activities

Field	Type	Meaning
ActivityId	INT, identity, primary key	Unique id
ActivityType	NVARCHAR(20)	Call, Email, Meeting, Task
Subject	NVARCHAR(200)	Short title
Notes	NVARCHAR(MAX)	Details
DueDate	DATETIME2	When it is due
IsCompleted	BIT	Done or not
LeadId, AccountId, ContactId, OpportunityId	INT, nullable foreign keys	What it relates to
CreatedAt	DATETIME2	Audit timestamp

Part 3: SQL script (run in SSMS)

Open SSMS, connect to AYAN-RIYAN, click New Query, and paste the script below. Run it once. It is safe to rerun, because each table is only created if it does not already exist.

```
USE agentic_db;
GO

-- ACCOUNTS
IF OBJECT_ID('dbo.Accounts', 'U') IS NULL
CREATE TABLE dbo.Accounts (
    AccountId      INT IDENTITY(1,1) PRIMARY KEY,
    AccountName    NVARCHAR(200) NOT NULL,
    Industry       NVARCHAR(100) NULL,
    Website        NVARCHAR(200) NULL,
    Phone          NVARCHAR(30)  NULL,
    City           NVARCHAR(100) NULL,
    Country        NVARCHAR(100) NULL,
    AnnualRevenue  DECIMAL(18,2) NULL,
    CreatedAt      DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt      DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME()
);
GO

-- CONTACTS
IF OBJECT_ID('dbo.Contacts', 'U') IS NULL
CREATE TABLE dbo.Contacts (
    ContactId   INT IDENTITY(1,1) PRIMARY KEY,
    AccountId   INT NOT NULL,
    FirstName   NVARCHAR(100) NOT NULL,
    LastName    NVARCHAR(100) NOT NULL,
    Email       NVARCHAR(200) NULL,
    Phone       NVARCHAR(30)  NULL,
    JobTitle    NVARCHAR(100) NULL,
    CreatedAt   DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt   DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT FK_Contacts_Accounts FOREIGN KEY (AccountId) REFERENCES
dbo.Accounts(AccountId),
    CONSTRAINT UQ_Contacts_Email UNIQUE (Email)
);
GO

-- LEADS
IF OBJECT_ID('dbo.Leads', 'U') IS NULL
CREATE TABLE dbo.Leads (
    LeadId                 INT IDENTITY(1,1) PRIMARY KEY,
    FirstName              NVARCHAR(100) NOT NULL,
    LastName               NVARCHAR(100) NOT NULL,
```

```
    Company                NVARCHAR(200) NULL,
    Email                  NVARCHAR(200) NULL,
    Phone                  NVARCHAR(30)  NULL,
    LeadSource             NVARCHAR(50)  NULL,
    Status                 NVARCHAR(30)  NOT NULL DEFAULT 'New',
    IsConverted            BIT NOT NULL DEFAULT 0,
    ConvertedAccountId     INT NULL,
    ConvertedContactId     INT NULL,
    ConvertedOpportunityId INT NULL,
    CreatedAt              DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt              DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT CK_Leads_Status CHECK (Status IN
('New','Contacted','Qualified','Unqualified','Converted'))
);
GO

-- OPPORTUNITIES
IF OBJECT_ID('dbo.Opportunities', 'U') IS NULL
CREATE TABLE dbo.Opportunities (
    OpportunityId    INT IDENTITY(1,1) PRIMARY KEY,
    AccountId        INT NOT NULL,
    ContactId        INT NULL,
    OpportunityName  NVARCHAR(200) NOT NULL,
    Stage            NVARCHAR(30)  NOT NULL DEFAULT 'Prospecting',
    Amount           DECIMAL(18,2) NULL,
    Probability      INT NULL,
    CloseDate        DATE NULL,
    CreatedAt        DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt        DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT FK_Opps_Accounts FOREIGN KEY (AccountId) REFERENCES
dbo.Accounts(AccountId),
    CONSTRAINT FK_Opps_Contacts FOREIGN KEY (ContactId) REFERENCES
dbo.Contacts(ContactId),
    CONSTRAINT CK_Opps_Stage CHECK (Stage IN
('Prospecting','Qualification','Proposal','Negotiation','Closed Won','Closed
Lost')),
    CONSTRAINT CK_Opps_Prob CHECK (Probability BETWEEN 0 AND 100)
);
GO

-- ACTIVITIES
IF OBJECT_ID('dbo.Activities', 'U') IS NULL
CREATE TABLE dbo.Activities (
```

```
    ActivityId     INT IDENTITY(1,1) PRIMARY KEY,
    ActivityType   NVARCHAR(20) NOT NULL,
    Subject        NVARCHAR(200) NOT NULL,
    Notes          NVARCHAR(MAX) NULL,
    DueDate        DATETIME2 NULL,
    IsCompleted    BIT NOT NULL DEFAULT 0,
    LeadId         INT NULL,
    AccountId      INT NULL,
    ContactId      INT NULL,
    OpportunityId  INT NULL,
    CreatedAt      DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT FK_Act_Leads FOREIGN KEY (LeadId) REFERENCES
dbo.Leads(LeadId),
    CONSTRAINT FK_Act_Accounts FOREIGN KEY (AccountId) REFERENCES
dbo.Accounts(AccountId),
    CONSTRAINT FK_Act_Contacts FOREIGN KEY (ContactId) REFERENCES
dbo.Contacts(ContactId),
    CONSTRAINT FK_Act_Opps FOREIGN KEY (OpportunityId) REFERENCES
dbo.Opportunities(OpportunityId),
    CONSTRAINT CK_Act_Type CHECK (ActivityType IN
('Call','Email','Meeting','Task'))
);
GO
```

Part 4: Sample data

Run this second script so your statistics tool has something to analyse.

```
USE agentic_db;
GO

INSERT INTO dbo.Accounts (AccountName, Industry, Website, Phone, City,
Country, AnnualRevenue) VALUES
('Acme Technologies', 'Technology', 'www.acmetech.com', '+91-9800000001',
'Kolkata', 'India', 5000000),
('Bright Retail Pvt Ltd', 'Retail', 'www.brightretail.in', '+91-9800000002',
'Pune', 'India', 2500000),
('Nova Healthcare', 'Healthcare', 'www.novahealth.com', '+91-9800000003',
'Bengaluru', 'India', 8000000);

INSERT INTO dbo.Contacts (AccountId, FirstName, LastName, Email, Phone,
JobTitle) VALUES
```

```
(1, 'Riya', 'Sen', 'riya.sen@acmetech.com', '+91-9811111111', 'CTO'),
(2, 'Amit', 'Shah', 'amit.shah@brightretail.in', '+91-9822222222',
'Procurement Head'),
(3, 'Neha', 'Rao', 'neha.rao@novahealth.com', '+91-9833333333', 'IT
Director');

INSERT INTO dbo.Leads (FirstName, LastName, Company, Email, Phone,
LeadSource, Status) VALUES
('Karan', 'Mehta', 'Zenith Logistics', 'karan@zenithlog.com', '+91-
9844444444', 'Web', 'New'),
('Sneha', 'Das', 'Pixel Studio', 'sneha@pixelstudio.in', '+91-9855555555',
'Referral', 'Contacted'),
('Rahul', 'Verma', 'Orbit Foods', 'rahul@orbitfoods.com', '+91-9866666666',
'Event', 'Qualified');

INSERT INTO dbo.Opportunities (AccountId, ContactId, OpportunityName, Stage,
Amount, Probability, CloseDate) VALUES
(1, 1, 'Acme CRM Rollout', 'Proposal', 120000, 60, '2026-12-15'),
(2, 2, 'Bright Retail POS Integration', 'Qualification', 45000, 30, '2027-01-
20'),
(3, 3, 'Nova Patient Portal', 'Negotiation', 200000, 80, '2026-11-30');

INSERT INTO dbo.Activities (ActivityType, Subject, Notes, DueDate, LeadId,
AccountId, ContactId, OpportunityId) VALUES
('Call', 'Intro call with Karan', 'Discuss logistics needs', '2026-11-05', 1,
NULL, NULL, NULL),
('Meeting', 'Proposal review with Acme', 'Walk through pricing', '2026-11-
10', NULL, 1, 1, 1),
('Email', 'Send contract to Nova', 'Final terms attached', '2026-11-08',
NULL, 3, 3, 3);
GO
```

Part 5: Verify in SSMS

```
USE agentic_db;
SELECT 'Accounts' AS TableName, COUNT(*) AS Rows FROM dbo.Accounts
UNION ALL SELECT 'Contacts', COUNT(*) FROM dbo.Contacts
UNION ALL SELECT 'Leads', COUNT(*) FROM dbo.Leads
UNION ALL SELECT 'Opportunities', COUNT(*) FROM dbo.Opportunities
UNION ALL SELECT 'Activities', COUNT(*) FROM dbo.Activities;
```

You should see 3 rows in each table.

Part 6: Connect it to your MCP server

Your existing server.py already works with these tables with no changes:

- `list_tables` will show all five tables.
- `describe_table` with `Leads` shows the columns.
- `fetch_rows` with `Opportunities` returns the sample deals.
- `table_statistics` with `Opportunities` gives min, max, and average of Amount and Probability.

Test all four in the MCP Inspector before moving on.

Part 7: The agentic roadmap (next steps)

1. **Read tools (done)**: list, describe, fetch, statistics.
2. **Targeted read tools**: `get_leads_by_status`, `get_open_opportunities`, `pipeline_summary_by_stage`.
3. **Write tools with validation**: `create_lead`, `create_account`, `create_contact`, `create_opportunity`. Each takes fixed parameters, validates them (required fields, allowed values for Status and Stage, email format), and uses parameterised INSERT statements. Never let the model run raw SQL.
4. **Workflow tool**: `convert_lead` which, inside one database transaction, creates the Account and Contact, creates the Opportunity, marks the lead as Converted, and stores the new ids on the lead. If any step fails, everything rolls back.
5. **Safety**: use a separate database login with only the permissions the tools need, log every write, and keep a confirmation step before conversions.
6. **Later**: rebuild the same server in C# with the official MCP SDK, once the concepts are clear.

Design notes

- Check constraints on Status, Stage, and ActivityType stop bad values at the database level, so an agent cannot save invalid data.
- Foreign keys keep the data connected: you cannot create a contact for an account that does not exist.

- The Leads table stores the converted ids, so you can always trace which lead became which deal.
- This is a starter design. Real CRMs add owners, territories, products, line items, and more, and we can grow it table by table.